

FLEET RULE OF THE ROAD TEST OCT 15

This test is based on: The International Regulations for Preventing Collisions at Sea (1972) with amendments in force up to December 2009 as detailed in "A Seaman's Guide Pocket Book of The International Regulations for Preventing Collisions at Sea" (BR 453 Supp), ATP1(F) Vol I and the Mariners Handbook (NP100).

The test is designed to be given to any Warfare Officer, Deck officer, Officer undertaking bridge watch-keeping duties or Young Officer/Cadet. The test should be administered under written exam conditions, the results should be recorded in the OOW training records for inspection by FLEET/FOST navigation staff. The exam should take no longer than 1 hr.

In this test, any change in the status of the ship or of the environment will be indicated in the script. Rules 1 to 19, 34, 35 and Annex IV must be known instinctively for instant application. A thorough knowledge of the other rules is required. The pass mark is 90%.

Answers are provided for uniformity of marking. This test follows the same format as the Ship Command Rule of the Road paper.

You are the OOW on RFA WAVE KNIGHT. You are deployed to the Caribbean on APT(N) duties and are scheduled for a port visit to Antigua. It is the morning watch, the weather is fine but heavy rain is forecast for later in the day.

1. Given the good visibility, the Rules of which section of which part apply? (R.4) (1)

The lookout reports a vessel at R45 displaying a black cylinder in her rigging. Several bearings show she is on a steady bearing.

2. What condition is this vessel claiming by displaying a black cylinder? (R.28) (1)

3. What lights (in addition to normal navigation lights) would she display by night to indicate this condition? (R.28) (1)

4. What relationship exists between these vessels and who should take action (R.18) (1)

5. What action (if any) should you take? (R.8 & 18) (2)

The range has reduced to only 2 miles and the bearing is still steady.

6. Who shall keep out of the way of whom now and why? (R.8 & 15) (2)

You pass the vessel safely and continue on your passage. The CO arrives on the Bridge and tells you he wants to quiz the Cadets later in the day on sound signals, lights and shapes. He suggests it may be a good idea for you to help them revise after your watch. After handing over the watch to the Second Officer, you head to the conference room with the Cadets and begin the revision. You pick on pilot vessels to begin with before moving onto anchored vessels.

7. What would you expect the Cadets to say a pilot vessel should display to indicate his status by day and by night? (R.29 & ICS) (2)

8. What would you expect the Cadets to say was the sound signal sounded by a pilot vessel stopped in the water, in restricted visibility? (R.35) (1)

9. What would you expect the Cadets to say were the shapes and lights your vessel should display whilst at anchor, by day and night? (R.30) (4)

10. What would you expect the Cadets to say was the sound signal your vessel would sound whilst at anchor in restricted visibility and at what interval? (R.35) (2)

Satisfied the Cadets know their stuff you send them to the CO's cabin and you retire for some well earned rest. Later you are back on the Bridge as OOW2 to support the OOW during a replenishment with HMS LANCASTER. As LANCASTER commences her fast backdown, the Yeoman pulls the RAM shapes up the mast to indicate the ship is restricted in her ability to manoeuvre.

11. Other than vessels engaged in the transfer of persons, provisions or cargo whilst underway, what other vessel can claim to be restricted in their ability to manoeuvre? (R.3) (5)

During the replenishment LANCASTER (as OCE) orders you to turn on navigation lights.

12. What do the Rules say about when navigation lights should be displayed? (R.20) (2)

Now fully dark and still pumping fuel, you sight the following lights at G30 range 6 miles.

13. What kind of vessel is it? (R.24) (2)

	W	
	W	
	W	
W		
	R	R

The bearing is steady and the range is decreasing.

14. Who has to keep clear of whom? (R.18) (1)

The range reduces to 4 miles and the bearing remains steady.

15. What action may be prudent at this time? (R.34) (1)

The range has further reduced and is now just 2 miles.

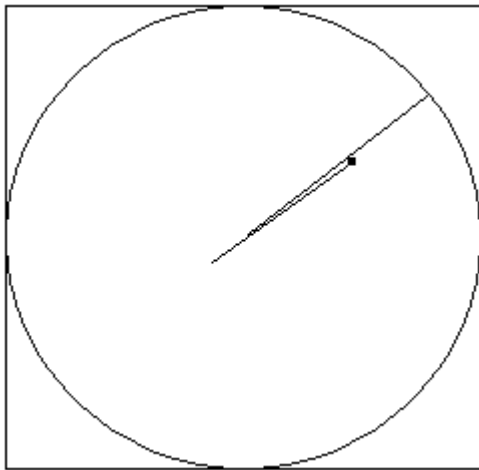
16. What action shall you take? (R.17) (1)

17. What do the Rules say the minimum range LANCASTER's (length 133m) stern light should be? (R.22) (1)

LANCASTER begins to disappear over the horizon to port when a heavy rainstorm engulfs her and moves towards you.

18. How do the Rules define restricted visibility? (R.3) (1)

Having lost sight of LANCASTER, you see on the radar a contact dead ahead at 6 miles. Nothing is visual because of the rain. You acquire the contact and after a few minutes you see the following relative vector (12 minutes) displayed:



19. What action (if any) do you take? (R.19) (3)

You pass that contact safely but don't see it because of the rain. You begin to contemplate what constitutes a safe speed in restricted visibility and reflect on the factors all vessels should take into account when determining a safe speed.

20. What are these factors all ships should take into account when determining a safe speed? (R.6) (6)

Just then the fog lookout reports hearing a sound signal from about R45 consisting of one prolonged blast followed by two shorts blasts.

21. What types of vessel could this be? (R.35) (3)

22. What action (if any) do you take? (R.19) (2)

The rain passes and the visibility improves to over 5Nm. You sight a vessel at G40.

23. What considerations will you take into account to determine if a risk of collision exists? (R.7)
(2)

24. If risk of collision exists, what action (if any) would you take? (R.15) (2)

25. Would a sound signal be appropriate? (R.34) (1)

Total _____ / 50

ANSWERS

You are the OOW on RFA WAVE KNIGHT. You are deployed to the Caribbean on APT(N) duties and are scheduled for a port visit to Antigua. It is the morning watch, the weather is fine but heavy rain is forecast for later in the day.

1. Given the good visibility, the Rules of which section of which part apply? (R.4) (1)

The Rules of Part B (Steering & Sailing Rules) Section I (Conduct of vessels in any condition of visibility) (1)

No marks for saying any other Part or any other Section (II is vessels in sight of one another, III is restricted visibility)

The lookout reports a vessel at R45 displaying a black cylinder in her rigging. Several bearings show she is on a steady bearing.

2. What condition is this vessel claiming by displaying a black cylinder? (R.28) (1)

She is claiming that she is constrained by her draught (1)

3. What lights (in addition to normal navigation lights) would she display by night to indicate this condition? (R.28) (1)

Three all round red lights in a vertical line (1)

4. What relationship exists between these vessels and who should take action (R.18) (1)

WVKN shall not impede the passage of the vessel CBD IAW R.18 (1)

5. What action (if any) should you take? (R.8f & 18) (2)

Keep clear (1) because:

(R.18d(ii)) Any vessel other than a vessel not under command or a vessel restricted in her ability to manoeuvre shall, if the circumstances of the case admit, avoid impeding the safe passage of a vessel constrained by her draught, exhibiting the signals in Rule 28.

and...

(R.8f(i)) A vessel which, by any of these Rules, is required not to impede the passage or safe passage of another vessel shall, when required by the circumstances of the case, take early action (½) to allow sufficient sea-room for the safe passage (½) of the other vessel.

Answers including slowing down (or stopping) and altering course to starboard (sounding one short blast) are acceptable but no marks for an alteration of course to port because of R.8f(ii).

The range has reduced to only 2 miles and the bearing is still steady.

6. Who shall keep out of the way of whom now and why? (R.8f & 15) (2)

The CBD vessel has an obligation to keep clear (1) under R.15 because:

(R.8f(iii)) A vessel the passage of which is not to be impeded remains fully obliged to comply with the Rules of this part [i.e. R.15] when the two vessels are approaching one another so as to involve risk of collision.

WVKN remains fully obliged not to impede the CBD vessel throughout (1).

You pass the vessel safely and continue on your passage. The CO arrives on the Bridge and tells you he wants to quiz the Cadets later in the day on sound signals, lights and

shapes. He suggests it may be a good idea for you to help them revise after your watch. After handing over the watch to the Second Officer, you head to the conference room with the Cadets and begin the revision. You pick on pilot vessels to begin with before moving onto anchored vessels.

7. What would you expect the Cadets to say a pilot vessel should display to indicate his status by day and by night? (R.29 & ICS) (2)

By day – Flag Hotel (1)

By night – two all-round lights in a vertical line, the upper being white and the lower red (1)

8. What would you expect the Cadets to say was the sound signal sounded by a pilot vessel stopped in the water, in restricted visibility? (R.35) (1)

Two prolonged blasts followed by four short blasts (1) because:

A pilot vessel when engaged on pilotage duty may in addition to the signals prescribed in paragraphs (a), (b) or (g) of this Rule sound an identity signal of four short blasts and...

(R.35b) A power-driven vessel underway but stopped and making no way through the water shall sound at intervals of not more than 2 minutes two prolonged blasts in succession with an interval of about 2 seconds between them.

9. What would you expect the Cadets to say were the shapes and lights your vessel should display whilst at anchor, by day and night? (R.30) (4)

A vessel at anchor shall exhibit where it can best be seen:

(i) in the fore part, an all-round white light (1) or one ball (1)

(ii) at or near the stern and at a lower level than the light prescribed in subparagraph (i), an all-round white light (1)

A vessel at anchor may, and a vessel of more than 100m in length shall, also use the available working or equivalent lights to illuminate her decks (1).

10. What would you expect the Cadets to say was the sound signal your vessel would sound whilst at anchor in restricted visibility and at what interval? (R.35) (2)

5 seconds rapid ringing of the bell in the fore part of the vessel followed immediately by 5 seconds rapid sounding of the gong in the aft part (1) at intervals not exceeding 1 minute (1).

Satisfied the Cadets know their stuff you send them to the CO's cabin and you retire for some well earned rest. Later you are back on the Bridge as OOW2 to support the OOW during a replenishment with HMS LANCASTER. As LANCASTER commences her fast backdown, the Yeoman pulls the RAM shapes up the mast to indicate the ship is restricted in her ability to manoeuvre.

11. Other than vessels engaged in the transfer of persons, provisions or cargo whilst underway, what other vessel can claim to be restricted in their ability to manoeuvre? (R.3) (5)

Vessels engaged in laying, servicing or picking up a navigational mark, submarine pipeline...(1)

Vessels engaged in surveying, dredging or other underwater operations (1)

Vessels engaged in mine clearance operations (1)

Vessels engaged in launching and recovering aircraft (1)

Vessels engaged in a towing operation where the tow severely restricts the towing vessel in her ability to deviate from the course she is following (1)

During the replenishment LANCASTER (as OCE) orders you to turn on navigation lights.

12. What do the Rules say about when navigation lights should be displayed? (R.20) (2)

**(R.20b) The Rules concerning lights shall be complied with from sunset and sunrise ($\frac{1}{2}$), and during such times no other lights shall be exhibited, except such lights as cannot be mistaken for the lights specified in these Rules or do not impair their visibility or distinctive character, or interfere with the keeping of a proper lookout ($\frac{1}{2}$).
(R.20c) The lights prescribed by these Rules shall, if carried, also be exhibited from sunrise to sunset in restricted visibility ($\frac{1}{2}$) and may be exhibited in all other circumstances when it is deemed necessary ($\frac{1}{2}$).**

Now fully dark and still pumping fuel, you sight the following lights at G30 range 6 miles.

13. What kind of vessel is it? (R.24) (2)

W
W
W
W
R R

Power-driven vessel ($\frac{1}{2}$) probably greater than 50m in length ($\frac{1}{2}$) engaged in towing ($\frac{1}{2}$) where the length of tow exceeds 200m ($\frac{1}{2}$)

The bearing is steady and the range is decreasing.

14. Who has to keep clear of whom? (R.18) (1)

**The towing vessel should keep clear of WAVE KNIGHT (1) because:
(R.18a) A power driven vessel underway shall keep clear of:
(ii) a vessel restricted in her ability to manoeuvre.**

The range reduces to 4 miles and the bearing remains steady.

15. What action may be prudent at this time? (R.34) (1)

**Sound 5 short rapid blasts (1) to indicate you are unclear on his intentions
or...
Saying you make take action when it has become clear that the give way vessel is not taking appropriate action in accordance with the Rules (R17d) is acceptable for the mark although this will require URG manoeuvring which may take time.**

The range has further reduced and is now just 2 miles.

16. What action shall you take? (R.17) (1)

Best action as will best aid to avoid collision (1). Clearly this will mean URG manoeuvring (if initiated in time above) or an emergency breakaway.

The other vessel passes safely and you complete the replenishment. The lookout reports that as LANCASTER peels away to port her stern light is not very bright.

17. What do the Rules say the minimum range LANCASTER's (length 133m) stern light should be? (R.22) (1)

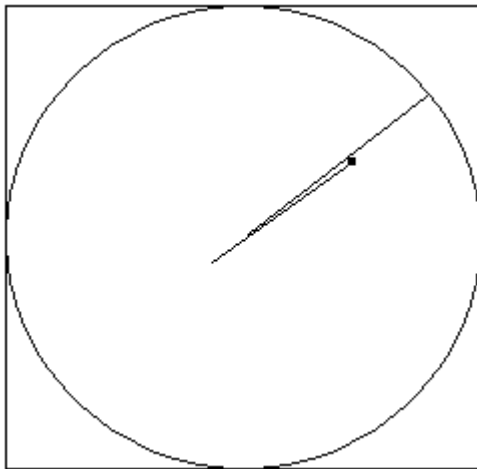
3 miles (1)

LANCASTER begins to disappear over the horizon to port when a heavy rainstorm engulfs her and moves towards you.

18. How do the Rules define restricted visibility? (R.3) (1)

The term restricted visibility means any condition where the visibility is reduced by fog, mist, heavy, sandstorms, snowstorms or any other similar cause (1)

Having lost sight of LANCASTER, you see on the radar a contact dead ahead at 6 miles. Nothing is visual because of the rain. You acquire the contact and after a few minutes you see the following relative vector (12 minutes) displayed:



19. What action (if any) do you take? (R.19) (3)

A vessel which detects by radar alone the presence of another vessel shall determine if a close-quarters situation is developing and/or risk of collision exists (1). If so, she shall take avoiding action in ample time, provided that when such action consists of an alteration of course, so far as possible the following shall be avoided:

- (i) an alteration of course to port for a vessel forward of the beam other than for a vessel being overtaken (1);**
- (ii) an alteration of course towards a vessel abeam or abaft the beam (1).**

You pass that contact safely but don't see it because of the rain. You begin to contemplate what constitutes a safe speed in restricted visibility and reflect on the factors all vessels should take into account when determining a safe speed.

20. What are these factors all ships should take into account when determining a safe speed? (R.6) (6)

**The state of visibility (1);
The traffic density including concentrations of fishing vessels or any other vessels (1);
The manoeuvrability of the vessel with special reference to stopping distance and turning ability in the prevailing conditions (1);
At night the presence of background light such as from shore lights or from backscatter of her own lights (1);
The state of wind, sea and current, and the proximity of navigational hazards (1);
The draught in relation to the available depth of water (1).**

Just then the fog lookout reports hearing a sound signal from about R45 consisting of one prolonged blast followed by two shorts blasts.

21. What types of vessel could this be? (R.35) (3)

(½ mark each)

RAM

NUC

Vessel engaged in fishing

Sailing vessel

Towing vessel

CBD

22. What action (if any) do you take? (R.19) (2)

Except where it has been determined that risk of collision does not exist, every vessel which hears apparently forward of her beam the fog signal of another vessel, or which cannot avoid a close-quarters situation with another vessel forward of her beam, shall reduce her speed to the minimum at which she can be kept on her course (1). She shall if necessary take all her way off and in any event navigate with extreme caution until danger of collision is over (1).

The rain passes and the visibility improves to over 5Nm. You sight a vessel at G40.

23. What considerations will you take into account to determine if a risk of collision exists? (R.7) (2)

In determining if risk of collision exists the following considerations shall be among those taken into account:

(i) such risk shall be deemed to exist if the compass bearing of an approaching vessel does not appreciably change (1);

(ii) such risk may sometimes exist even when an appreciable bearing change is evident, particularly when approaching a very large vessel or a tow or when approaching a vessel at close range (1).

24. If risk of collision exists, what action (if any) would you take? (R.15) (2)

When two power-driven vessels are crossing so as to involve risk of collision, the vessel which has the other on her own starboard side shall keep out of the way (1) and shall, if the circumstances of the case admit, avoid crossing ahead of the other vessel (1).

25. Would a sound signal be appropriate? (R.34) (1)

(R.34a): one short blast (1) to mean “I am altering my course to starboard”

Total _____ / 50